



Week 6 Lecture 1

▼ Class	BSCCS2003
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🔗 Materials	
# Module #	32
▼ Type	Lecture
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API Design

Distributed Software Architecture

- Server - Clients
- Standard "protocols" needed for communication
- Assumptions?
 - Server always on?
 - Server knows what client is doing?
 - Client authentication
 - Network latency?

The Web

- Client - Server may be far apart
- Different network, latencies, quality
- Authentication? Not core part of protocol
- State?
 - Server does not know what state client is in
 - Client cannot be sure what state server is in

Architecture for the Web

- Roy Fielding, PhD thesis 2000 UC Irvine

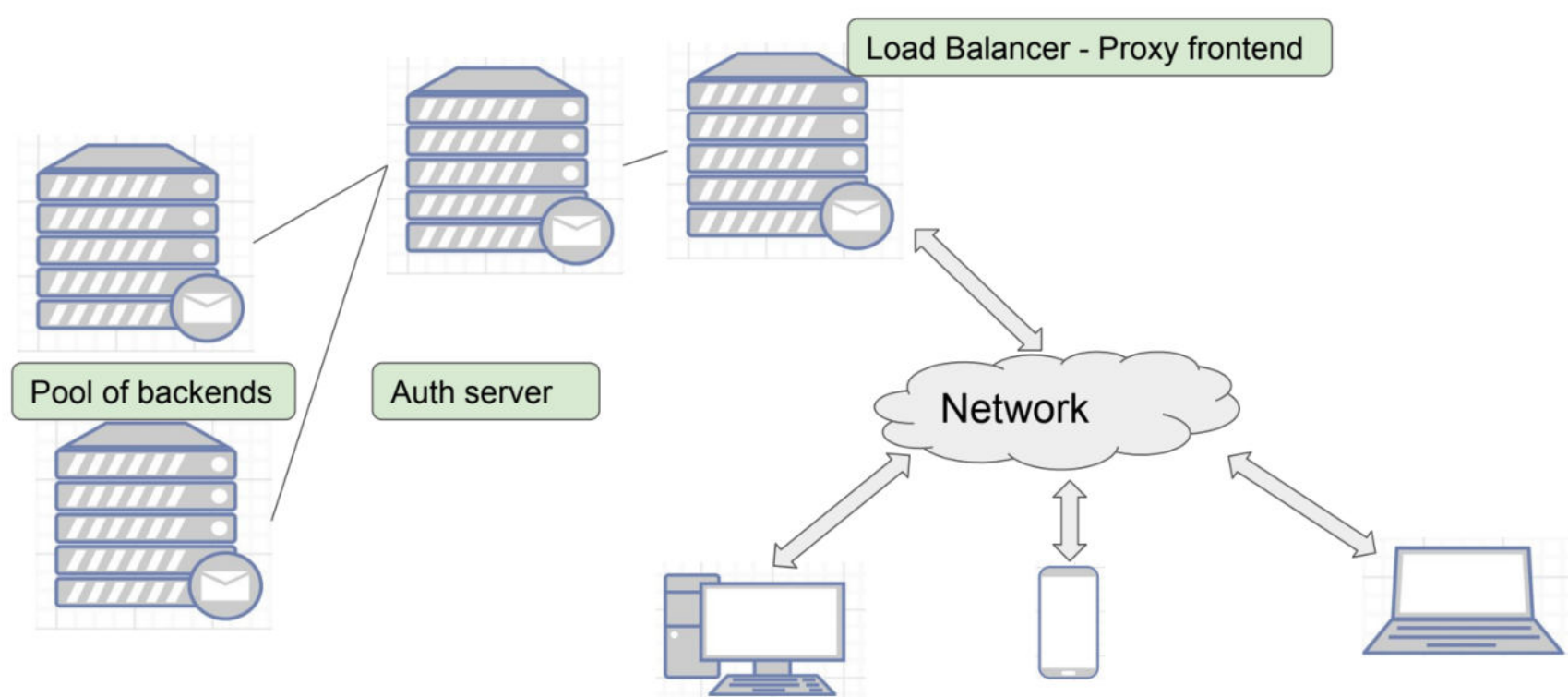
- "REpresentational State Transfer" - REST
 - Take into account the limitations of the web
 - Provide guidelines or constraints
- Software Architecture Style
 - Not a set of rules

Constraint #1: Client-Server

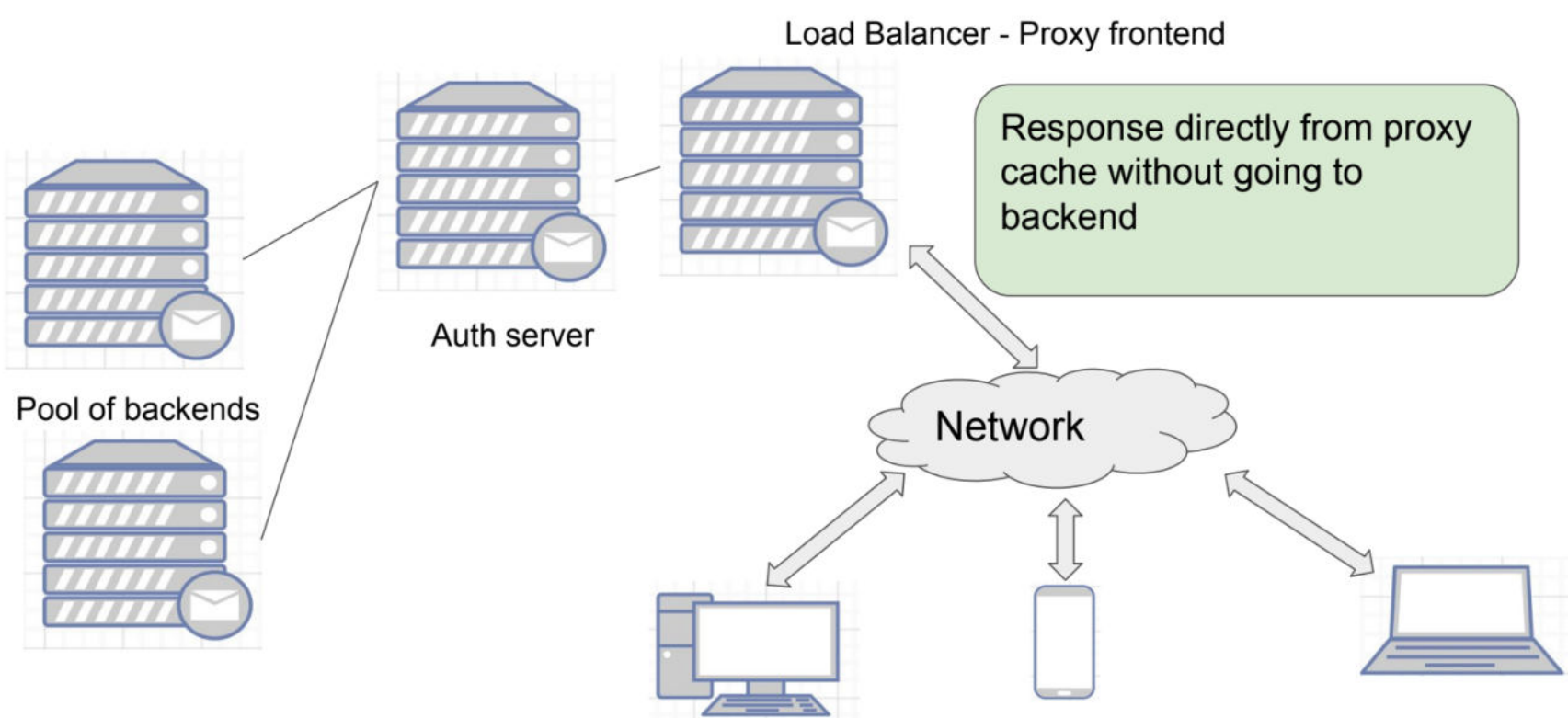
Constraint #2: Stateless

- Server cannot assume state of the client:
 - Which page are you looking at
 - Is a request coming from an already logged in user just because of the address?
- Client cannot assume the state of the server
 - Did the server reboot since the last request?
 - Is this request being answered by the same server?

Constraint #3: Layered system



Constraint #4: Cacheability



Constraint #5: Uniform interface

- Client and State interact in a uniform and predictable manner
- Server exposes "resources"

Hypertext/media used to convey the available resources and functionality - can be discovered by client through hypertext information from the server

Constraint #6: Code on Demand (Optional)

- Server can extend client functionality
 - JavaScript
 - Java applets (are they still relevant?)

Part of the overall structure - these are not hard rules